

## Towards causally aware sensitivity analysis?

# Motivation

## The correlation pitfalls

Sobol indices and Shapley values are correlation based

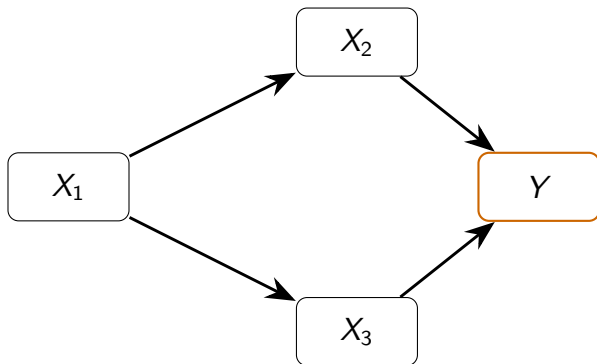
- ▶ A feature can look important simply because it is a proxy for the true cause
- ▶ Marginal and conditional Shapley values cannot separate a feature's own effect from what it only mediates.

# Motivation

## The correlation pitfalls

- ▶ Toy structural model:  $X_1$  drives  $X_2$  and  $X_3$ , which both feed  $Y$  directly
- ▶  $X_1$  itself has **no direct edge into**  $Y$ .

## Motivation



See `Lecture2a.ipynb`

# Motivation

## Towards actionable explanations

- ▶ Aim : go beyond observational setting
- ▶ Tells you what to change
- ▶ Gives you a valid guarantee about what will happen if you do.  
That's two separate requirements, and most explanation methods satisfy at most one:

# Motivation

## Mutability

- ▶ The explanation must point to a variable an agent can actually manipulate.
- ▶ "Age" can be causally important without being actionable
- ▶ Recourse requires identifying feasible changes, not just important ones

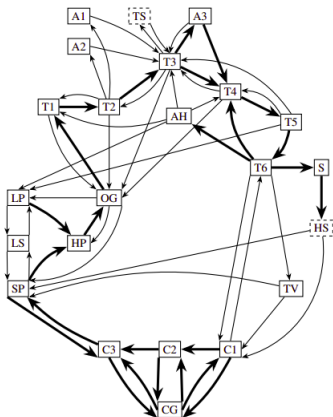
# Motivation

## Quantifying an action effect

- ▶ If a feature is flagged as important, acting on it should have an important effect
- ▶ This is not measured with correlation
- ▶ Towards **causally aware** importance measures?

## Why causality?

A motivating dataset from the European Spallation Source (ESS)



## Motivation

- ▶ The data was generated by the ACCP system (a refrigeration system),
- ▶ A number of process variables (PVs) describing the state of the ACCP were measured over time.
- ▶ Observations from three separate time intervals are available (Period 1, 2, and 3).
- ▶ Each PV belongs to a subsystem, and each subsystem is a physical component of the ACCP system (such as a compressor).

## Motivation

| Index | Description      | No. of PVs | Index | Description              | No. of PVs |
|-------|------------------|------------|-------|--------------------------|------------|
| T1    | Turbine 1        | 5          | C1    | cold Compressor 1        | 5          |
| T2    | Turbine 2        | 5          | C2    | cold Compressor 2        | 6          |
| T3    | Turbine 3        | 8          | C3    | cold Compressor 3        | 6          |
| T4    | Turbine 4        | 5          | CG    | Compressor General       | 11         |
| T5    | Turbine 5        | 6          | SP    | Sub-atm. Prs. compressor | 17         |
| T6    | Turbine 6        | 8          | SPWE  | SP line Warm End         | 2          |
| TG    | Turbine General  | 1          | LP    | Low Pressure compressor  | 16         |
| A1    | Adsorber 1 80 K  | 3          | LPWE  | LP line Warm End         | 2          |
| A2    | Adsorber 2 80 K  | 4          | LS    | LP+SP compressor         | 8          |
| A3    | Adsorber 3 20 K  | 4          | MPWE  | MP line Warm End         | 3          |
| HX1   | Heat eXchanger 1 | 2          | HP    | High Pressure compressor | 22         |
| HX2   | Heat eXchanger 2 | 9          | HPWE  | HP line Warm End         | 5          |
| HX3   | Heat eXchanger 3 | 2          | OG    | Oil and Gas removal      | 15         |
| HX4   | Heat eXchanger 4 | 3          | AH    | Ambient Heater           | 5          |
| HX5   | Heat eXchanger 5 | 2          | D     | Dewar                    | 9          |
| HX6   | Heat eXchanger 6 | 5          | S     | Subcooler                | 7          |
| TV    | Test Vessel      | 8          | TS    | Thermal Shield*          | 8          |
| HS    | Helium Supply*   | 6          |       |                          |            |

# Motivation

## Practical questions

- ▶ Can we intervene to control the output of the system?
- ▶ Can we prevent derives/anomalies?
- ▶ Can we identify the root causes of such derives/anomalies and derive a strategy of remediation

Causal modeling : a gold road to answer such questions?

# What is causality?

## Causal and non causal questions

- ▶ Non-causal: investigate purely statistical associations
- ▶ Causal effects : Estimating causal effects (of hypothetical interventions) using the causal graph
- ▶ Counterfactuals : Such as attribution of observed events or mediation analysis

Caveat : the graph **could be unknown** and in this case we have to estimate it!

## First scientific question : causal discovery

What about the case where the DAG is unknown?

- ▶ In general, causal discovery from observational data is not possible.
- ▶ But it is possible under additional assumptions.

## Second scientific question : causal inference, effect of an intervention

### Causal quantities

- ▶ Assume now DAG known or estimated
- ▶ We cannot quantify the output of an intervention for a given individual
- ▶ A causal quantity

$$ATE = \mathbb{E}[Y|do(T = 1)] - \mathbb{E}[Y|do(T = 0)]$$

## Second scientific question : causal inference, effect of an intervention

### Scientific questions

- ▶ Practical estimation of the causal effects from (observational) data?
- ▶ Do calculus, identifiability questions

See `Lecture2b.ipynb`